

Number theory – 8

Pell equations, Markov-type equations and Vieta jumping

Homework

24 сентября 2026 года

Theoretical minimum. If $|\alpha - p/q| < 1/(2q^2)$, then p/q is a convergent of α (Legendre); so a solution of $0 < |x^2 - dy^2| < \sqrt{d}$ comes from a convergent of $\sqrt{d}$. Continued fractions are eventually periodic exactly for quadratic irrationals, $\sqrt{d} = [a_0; \overline{a_1, \dots, a_{r-1}, 2a_0}]$, and $p_{k-1}^2 - dq_{k-1}^2 = (-1)^k Q_k$ with $0 < Q_k < 2\sqrt{d}$, $Q_k = 1$ iff $r \mid k$. The norm $x^2 - dy^2$ is multiplicative, the units of $\mathbb{Z}[\sqrt{d}]$ are $\pm\eta^k$, the solutions of $x^2 - dy^2 = 1$ are $\pm\varepsilon^k$ with $\varepsilon \in \{\eta, \eta^2\}$ (existence: Number theory – 5). For $n \neq 0$ the solutions of $x^2 - dy^2 = n$ form $h < \infty$ classes $\pm\beta\varepsilon^k$, and there are $\frac{h}{\ln \varepsilon} \ln X + O(1)$ of them with $0 < x \leq X$. Vieta jumping: with z also $z' = s - z = t/z$ is a root of $z^2 - sz + t = 0$; by descent all Markov triples $(x^2 + y^2 + z^2 = 3xyz)$ come from $(1, 1, 1)$ and form a binary tree, about $C(\ln N)^2$ of them below N , and the Hurwitz equation $\sum x_i^2 = a \prod x_i$ has finitely many fundamental solutions, those with $x_n \leq \frac{a}{2} x_1 \cdots x_{n-1}$.

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1. Find a triple of integers x, y, z , each greater than 50, satisfying $x^2 + y^2 + z^2 = 3xyz$. (VJIMC 1994)
 2. Prove that there exist infinitely many integers n such that $n, n+1, n+2$ are each the sum of two squares of integers. (Example: $0 = 0^2 + 0^2$, $1 = 0^2 + 1^2$, $2 = 1^2 + 1^2$.) (Putnam 2000)
 3. Evaluate

$$\sqrt[8]{2207 - \frac{1}{2207 - \frac{1}{2207 - \dots}}}$$

Express your answer in the form $\frac{a+b\sqrt{c}}{d}$, where a, b, c, d are integers.

(Putnam 1995)

4. Let $a_0 = 1$, $a_1 = 2$, and $a_n = 4a_{n-1} - a_{n-2}$ for $n \geq 2$. Find an odd prime factor of a_{2015} . (Putnam 2015)
5. Consider the power series expansion $\frac{1}{1-2x-x^2} = \sum_{n=0}^{\infty} a_n x^n$. Prove that, for each integer $n \geq 0$, there is an integer m such that $a_n^2 + a_{n+1}^2 = a_m$. (Putnam 1999)
6. Let $a_0 = 1$ and $a_{n+1} = \frac{7a_n + \sqrt{45a_n^2 - 36}}{2}$ for $n \geq 0$. Show that for all positive integers n : a) a_n is a positive integer; b) $a_n a_{n+1} - 1$ is the square of an integer. (VJIMC 2019)
7. Prove that the numbers of a Markov triple are pairwise coprime, that no Markov number is divisible by 4, and that every odd prime divisor of a Markov number is $\equiv 1 \pmod{4}$. Then prove that $x^2 + y^2 + z^2 = kxyz$ has a solution in positive integers only for $k = 1$ and $k = 3$, and that the solutions for $k = 1$ are exactly the triples $(3x, 3y, 3z)$ with (x, y, z) a Markov triple.
8. Let $d > 1$ be a non-square, $\varepsilon > 1$ the least solution of $x^2 - dy^2 = 1$ and $n \neq 0$. Prove that every solution of $x^2 - dy^2 = n$ equals $\pm \beta \varepsilon^k$ with $k \in \mathbb{Z}$ and $\beta = u + v\sqrt{d}$ a solution with $\sqrt{|n|} \leq \beta < \varepsilon \sqrt{|n|}$, and give explicit bounds for $|u|, |v|$. Deduce that the number of solutions in positive integers with $x \leq X$ is $\frac{h}{\ln \varepsilon} \ln X + O(1)$, where h is the number of classes. Find all solutions of $x^2 - 3y^2 = 13$ in positive integers.
9. Prove that the continued fraction of a real α is eventually periodic if and only if α is a quadratic irrational, and purely periodic if and only if, moreover, $\alpha > 1$ and its conjugate lies in $(-1, 0)$ (Galois). Deduce that $\sqrt{d} = [a_0; \overline{a_1, \dots, a_{r-1}, 2a_0}]$ with $a_1, \dots, a_{r-1}$ symmetric, compute the continued fractions of $\sqrt{m^2 + 1}$ and $\sqrt{m^2 - 1}$ ($m \geq 2$ in the second case), and find the corresponding units η and ε .
10. For a positive integer N , let S_N be the number of pairs of integers $1 \leq a, b \leq N$ such that $(a^2 + a)(b^2 + b)$ is a perfect square. Prove that the limit $\ell = \lim_{N \rightarrow \infty} S_N/N$ exists and find its value. Is it true that $|S_N - \ell N| \leq N^{1/2025}$ for all sufficiently large N ? (IMC 2025 and shortlist)